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Semaglutide and tirzepatide effects on cardiovascular outcomes in people with overweight o r obesity in the real world (STEER)

Lauren Wilson, Zhenxiang Zhao, Victoria Divino, Matthew Bassan, Briain O Hartaigh, Kerem Ozer

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ABSTRACT

Aims: To evaluate real-world cardiovascular outcomes among patients with overweight or obe sity receiving incretin therapy.

Methods: A matched observational cohort study of 21,250 adults with established atheroscle rotic cardiovascular disease without diabetes.

Cohorts were balanced across baseline clinical characteristics, baseline statin usage, and prior cardiovascular interventions.

Primary outcomes evaluated included 3-point major adverse cardiovascular events comprising myocardial infarction, stroke, and mortality.

REAL-WORLD CARDIOVASCULAR FINDINGS

Primary Outcome: Incretin pharmacotherapy was associated with a statistically significant 29 percent reduction in 3-point major cardiovascular events.

Hazard ratio for the primary composite cardiovascular endpoint was 0.71 with 95 percent co nfidence intervals between 0.58 and 0.87.

Hospitalization for heart failure and coronary revascularization were reduced by 22 percen t across the 24-month follow-up window.

The cardiometabolic protective benefits remained evident across both normal glucose tolera nce and pre-diabetes baseline strata.

Weight reduction averaged 14.2 percent in patients adhering to physician-supervised lifest yle and medical management protocols.

DISCUSSION & TRANSLATIONAL RELEVANCE

This real-world study reinforces that modern incretin therapy exerts cardioprotective effe
cts extending beyond mere aesthetic weight loss.
Physician supervision ensured appropriate dose titration, gastrointestinal adverse event m
anagement, and treatment persistence.
Study Limitations: Retrospective electronic health registry design subject to residual con
founding; prescription fill adherence is indirect.
Cardiovascular protection was magnified when pharmacological therapy was combined with act
ive lifestyle habit restructuring.
Conclusion: Incretin-based metabolic intervention delivers significant cardiovascular risk
reductions under rigorous clinical oversight.